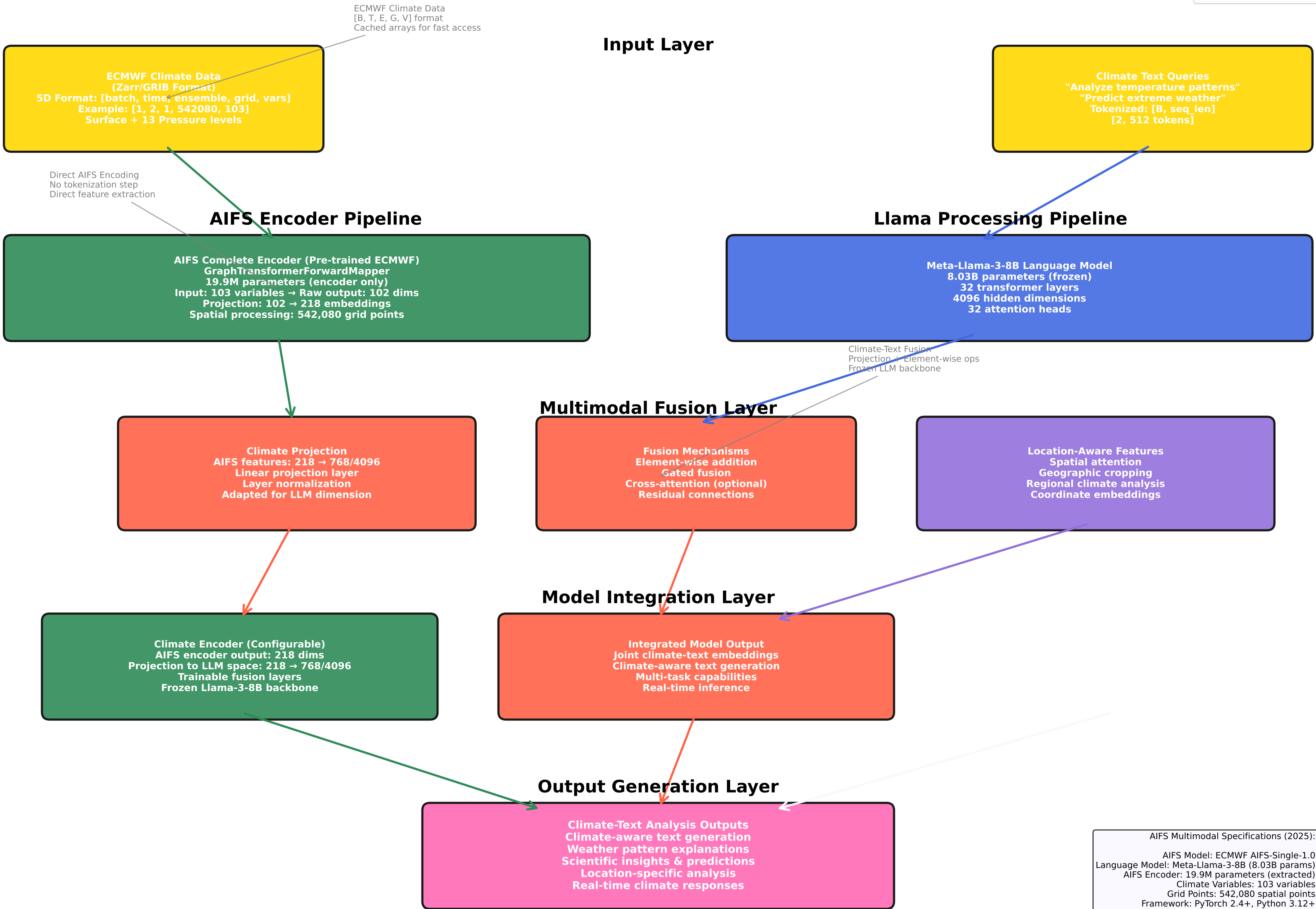


AIFS Multimodal Climate AI Architecture (2025)

ECMWF AIFS Encoder + Meta-Llama-3-8B Integration

- AIFS Components
- Llama Components
- Encoder/Processing
- Fusion Mechanisms
- Data Sources
- Outputs



AIFS Multimodal Specifications (2025):

- AIFS Model: ECMWF AIFS-Single-1.0
- Language Model: Meta-Llama-3-8B (8.03B params)
- AIFS Encoder: 19.9M parameters (extracted)
- Climate Variables: 103 variables
- Grid Points: 542,080 spatial points
- Framework: PyTorch 2.4+, Python 3.12+
- Training: CPU optimized, GPU compatible